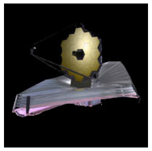

2074 - Galactic Archaeology with Brown Dwarfs in Medium-Deep JWST Observations

Cycle: 1, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

We propose an Archival study of brown dwarfs and subdwarfs in the North Ecliptic Pole Time-Domain Field that is taken as part of the guaranteed-time observations of Windhorst et al. While this GTO program is designed as an extragalactic survey, we propose to use the uniquely sensitive observations to identify populations of brown dwarfs in the Galactic halo and thin disk. This field will be observed in eight NIRCam imaging bands and one NIRISS wide-field slitless band, and has HST data from WFC3 and ACS. These filters and this spectroscopic mode are ideal for not only identifying substellar objects, but also for discriminating between the effects of surface gravity and metallicity. Since this is a medium-deep field at modest Galactic latitude (b~33 deg), we expect to find 20-30 brown dwarfs out to <14 kpc (or <7 kpc above the Galactic plane). These JWST data permit the first direct exploration of brown dwarfs in the Galactic halo, as current studies rely on identifying candidate halo brown dwarfs by their extreme proper motions. Therefore this survey is a unique opportunity to characterize subdwarfs and the coolest known substellar objects (Y-dwarfs). Cycle 1 archival support is critical to develop the tools to identify substellar objects in large JWST datasets on a timescale such that follow-up studies can proposed.

OBSERVING DESCRIPTION

This is an archival program, therefore we do not have an observing description. The data that we are requesting is eligible for archival programs, as it will be released publicly by the GTO team.